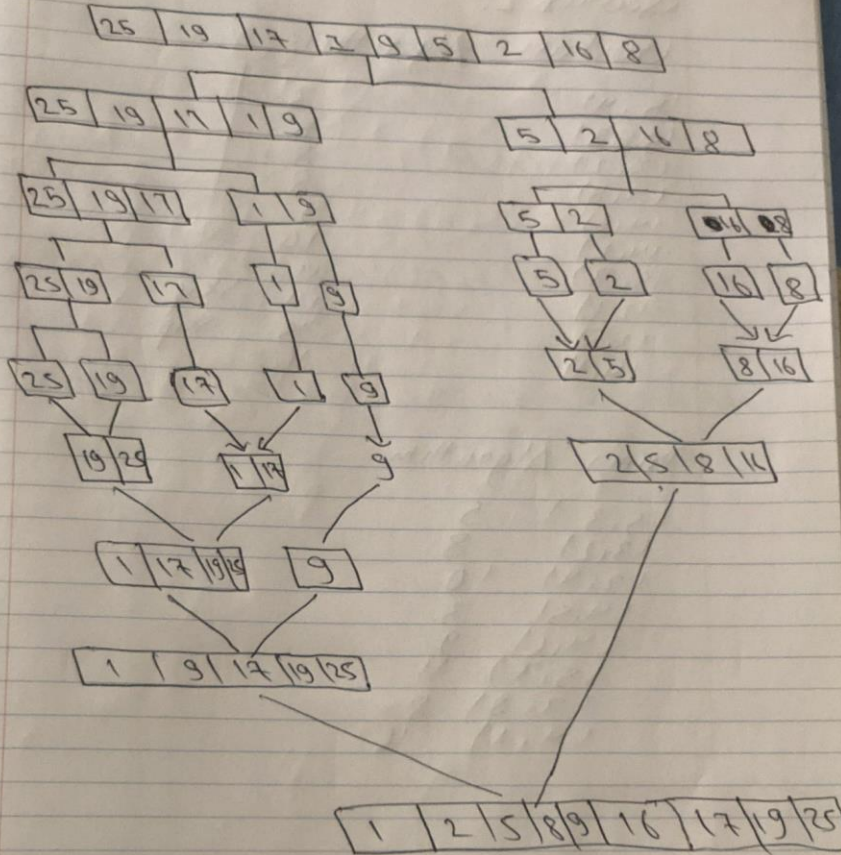


Home work -2

Q.2



Question 2:-

Kruskal's Algorithm

(5, 4)	3	
(6, 3)	5	
(3, 2)	8	
(7, 8)	8	
(4, 6)	9	
(4, 2)	10	
(5, 7)	11	
(2, 6)	14	Skip
(3, 5)	15	Skip
(4, 1)	16	

Prim's Algorithm

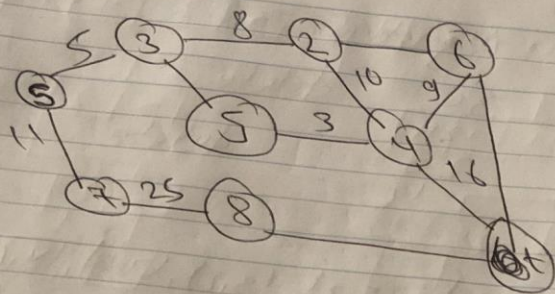
(5, 3)	5
(3, 2)	8
(2, 4)	10
(4, 5)	3
(4, 6)	9
(5, 7)	11
(7, 8)	8
(4, 1)	16

Reverse Algorithm

(8, 1)	30	delete
(7, 8)	25	
(6, 1)	20	delete
(9, 1)	16	

edges = (5, 4), (5, 3), (3, 2), (7, 8), (4, 6), (4, 2), (5, 7)
(4, 1)

$$\begin{aligned} \text{total weight} &= 3 + 5 + 8 + 25 + 9 + 10 \\ &\quad + 11 + 16 \\ &= 87 \end{aligned}$$



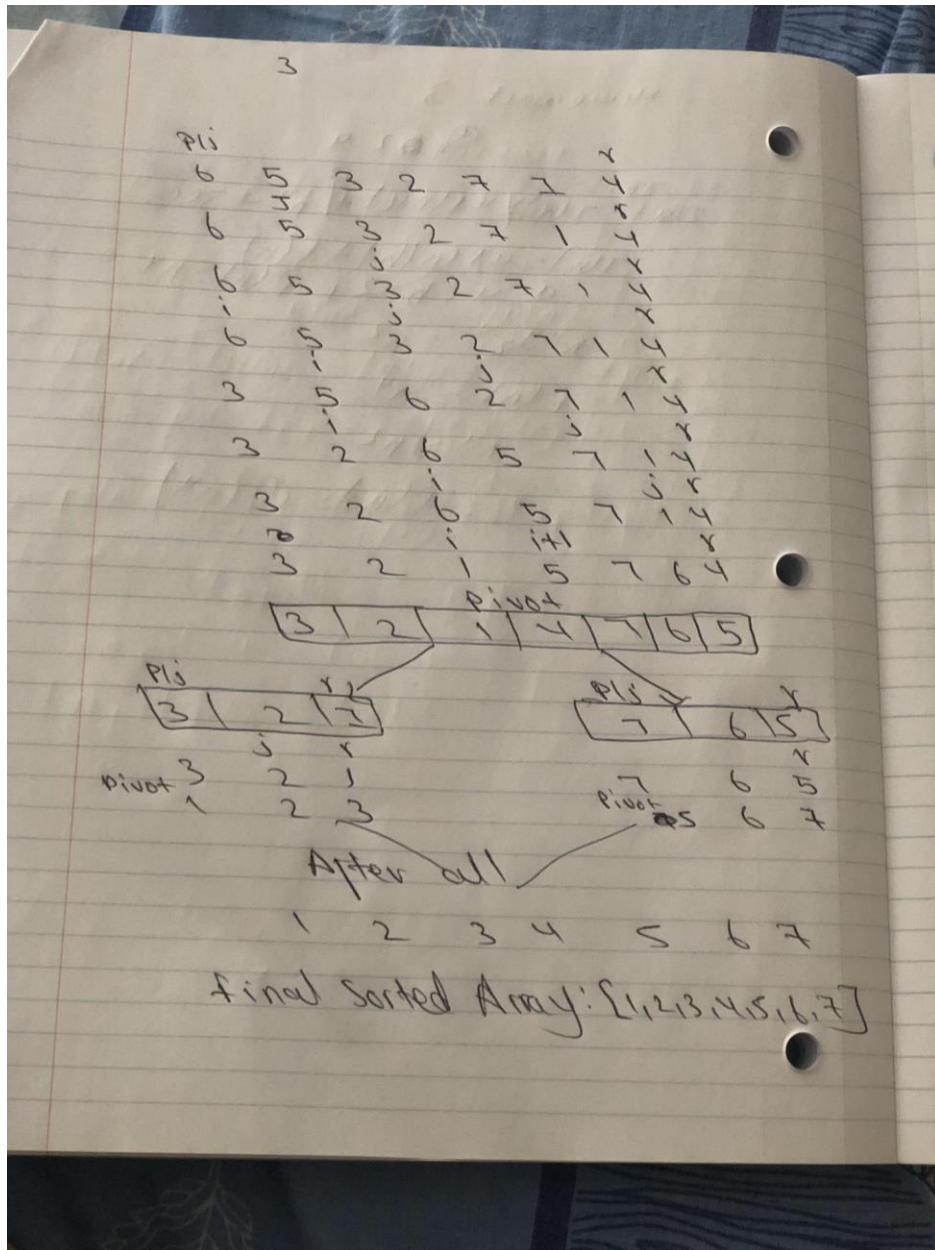
Q. 3.

Input $s = \{6, 5, 3, 2, 7, 1, 4\}$

Pivot: 4

j	A[j]	Action	i	Array state
-	-	Initial	-1	[6, 5, 3, 2, 7, 1, 4]
0	6	$6 > 4$, no swap	-1	[6, 5, 3, 2, 7, 1, 4]
1	5	$5 > 4$, no swap	-1	[6, 5, 3, 2, 7, 1, 4]
2	3	$3 < 4$, swap A[0], A[2]	0	[3, 5, 6, 2, 7, 1, 4]
3	2	$2 < 4$, swap A[0], A[3]	1	[2, 5, 6, 3, 7, 1, 4]
4	7	$7 > 4$, no swap	1	[2, 5, 6, 3, 7, 1, 4]
5	1	$1 < 4$, swap A[0], A[5]	2	[1, 5, 6, 3, 7, 2, 4]
-	-	final swap A[i], pivot	3	[1, 2, 5, 6, 3, 7, 4]

Pivot 4 is placed at index 3



This is just a images collection tha ti have made in my copy work